

The Protestant Ethic and the Principal-Agent Relation: Evidence from CEO Employment Agreements*

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Abstract

We ask whether the religious culture surrounding a firm's headquarters shapes the most basic formal instrument of the principal-agent relation: the written employment agreement between a firm and its chief executive. Using large language models to read exhibit filings from the SEC's EDGAR system, we assemble 6,373 explicit CEO employment contracts, of which 4,660 are fixed-term employment agreements (FEAs), covering 3,468 CEOs at 2,068 Execucomp firms between January 2005 and December 2022 — to our knowledge the most comprehensive sample of CEO employment contracts assembled to date. Firms headquartered in counties with higher Protestant adherence are markedly less likely to use FEAs, and the coefficient on Protestant adherence is negative and significant at the one percent level in every specification. Two alternative measures of contractual explicitness — an index built from ten key compensation provisions and the log length of the contract text — deliver the same negative relation, so agreements written in more Protestant counties are not merely rarer but thinner. The effect is denominational rather than devotional: total religious adherence carries no explanatory power, whereas Catholic adherence enters positively and significantly, so the two traditions are associated with contractual formality of opposite sign. Extending the analysis to forced CEO

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turnover yields directionally consistent but statistically insignificant evidence of greater board forbearance.

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“Seest thou a man diligent in his business? He shall stand before kings;”
— Bible, Proverbs 22:29

1 Introduction

In his monumental work in sociology and economic history, *The Protestant Ethic and the Spirit of Capitalism*, Max Weber (1930) posited that the Protestant ethic played a crucial role in the development of the capitalist economy. Central to Weber’s argument is the Protestant emphasis on approaching one’s worldly endeavors with a sense of responsibility and discipline, framed as a response to a “calling.” This perspective of elevating work to a divinely ordained task or vocation fostered a disciplined and rational approach to economic activity, marking a significant departure from the Catholic tradition, which placed greater importance on spiritual life and often regarded secular work as less central to one’s path to salvation.

While Max Weber examines the impact of the Protestant ethic on economic development through a sociological and historical lens, analyzing its effects at a broad societal level over an extended time scale, we adopt an empirical approach with a more focused lens: investigating how this ethic influences the quality of human collaboration in a principal-agent setting. Since Jensen and Meckling (1976) established that the separation of ownership and management in public firms leads to the agency cost, executive employment contracts have been extensively studied as key mechanisms to align the interests of a firm’s executives and its shareholders. In this context, we hypothesize that the Protestant tradition’s emphasis on work ethic, personal responsibility, and integrity in secular activities may help mitigate the relational friction arising from informational asymmetry within the principal-agent relationship, thereby influencing the design of the executive employment contracts.

From a contracting theory perspective, CEOs primarily invest their human capital, which is characterized by high specificity and non-transferability. This unique investment exposes CEOs to significant career risks, including potential future wage reductions,

reassignments, or termination, all of which can substantially impact their human capital value (Jensen and Meckling, 1976; Shleifer and Vishny, 1997).

To mitigate these risks, fixed-term employment agreements (FEAs) are often employed as a protective mechanism for CEO interests. These agreements provide a degree of job security and ensure predetermined compensation and status for a specified period, thereby incentivizing greater investment in firm-specific human capital (Gillan et al., 2009). However, recent studies suggest that informal governance mechanisms may also play a crucial role in shaping the CEO-firm relationship (Guiso et al., 2015).

This paper examines an alternative perspective on CEO employment agreements through the lens of Protestant work ethic. Building on Weber (1930)’s seminal work, we posit that the Protestant ethic, which emphasizes viewing work as a divine calling and encourages long-term commitment, may serve as an informal governance mechanism. We hypothesize that this ethical framework can potentially reduce the demand for strict FEAs by fostering an organizational culture based on trust, long-term orientation, and shared values.

We extend previous research on religious norms and corporate behavior (e.g., Hilary and Hui, 2009; Grullon et al., 2009) by focusing on how Protestant work ethic influences the contractual arrangements between firms and their CEOs. We observe a notable variation in contract structures depending on the religious composition of the firm’s geographic location. Specifically, we find that firms located in predominantly Protestant counties are less likely to use FEAs, and that the agreements they do write are shorter in text and contain fewer of the standard compensation provisions. The pattern is specific to Protestantism rather than to religiosity as such: overall religious adherence bears no significant relation to FEA use, while Catholic adherence is significantly *positively* related to it, so the two traditions are associated with contractual formality of opposite sign.

Our study examines the influence of Protestant social norms on CEO fixed-term employment agreement (FEA) choices across all firms in Execucomp from 2005 to 2022. We construct a comprehensive dataset by employing Large Language Models (LLMs)

to analyze EXHIBIT-10, EXHIBIT-99, and EXHIBIT-5 filings attached to various SEC forms, including 8-K, DEF-14A, 10-Q, 10-K, and S-4. This innovative approach enables efficient processing and classification of a vast array of SEC filings, substantially enhancing the breadth and precision of our data collection. Our final sample encompasses 6,373 explicit employment contracts (including 4,660 FEAs) executed between January 1, 2005 and December 31, 2022, covering 3,468 CEOs and 2,068 firms. To our knowledge, this dataset represents the most extensive and comprehensive sample of CEO employment contracts to date, surpassing previous studies that focused primarily on S&P 500 firms (Song and Wan, 2017; Gillan et al., 2009). By expanding the sample to all Execucomp firms and utilizing advanced text analysis techniques, our study offers a more exhaustive examination of CEO contract practices and their determinants across the U.S. public company landscape.

Our findings reveal distinct trends in explicit contract usage between S&P 500 and non-S&P 500 firms. S&P 500 companies demonstrate a relatively stable pattern, with explicit contract usage fluctuating around 40%, while non-S&P 500 firms consistently exhibit a higher adoption rate, averaging above 50%. However, both groups show a declining trend in FEA usage over time. For S&P 500 companies, FEA usage decreased from 37% in 2006 to 18% in 2022, while non-S&P 500 firms experienced a decline from 37% to 29% over the same period. Notably, non-S&P 500 firms consistently demonstrate a higher propensity to use both FEAs and explicit contracts. These results suggest that firm size and complexity, as proxied by S&P 500 membership, are associated with distinct approaches to CEO contracting. This divergence may reflect differences in bargaining power, governance structures, and labor market conditions between large, established firms and their smaller counterparts.

The results of our empirical investigation demonstrate a statistically significant negative association between Protestant adherence rates and the prevalence of FEA. Guiso et al. (2006) argue that cultural norms, including religious beliefs, can significantly influence trust and economic outcomes, which aligns with our finding that the Protestant ethic may mitigate concerns about opportunistic behavior and reduce incentives to violate

contract terms, fostering a more trusting environment between CEOs and firms.

To ensure the robustness of our results, we employ two additional measures of contract explicitness: the natural logarithm of length of each CEO’s contract and an explicit index based on ten key compensation provisions. This index, similar to the approach used by [Song and Wan \(2017\)](#), sums the presence of provisions such as base salary specification, target bonus clauses, and change in control clauses, providing a comprehensive measure of contract explicitness. These additional tests corroborate our main findings, strengthening the validity of our conclusions. Our comparative analyses with total religiosity and Catholic adherence rates find no similarly significant negative relationships; total adherence is statistically indistinguishable from zero throughout, while Catholic adherence enters positively and significantly. This denominational asymmetry indicates that the observed effect is not merely a function of general religiosity or of local risk aversion, but a specific manifestation of Protestant values — although the significantly positive Catholic coefficient also qualifies any simple application of [Weber \(1930\)](#)’s thesis, a point we return to in Section 4.3.

We finally set out a framework for examining how Protestant culture relates to CEO forced turnover decisions, particularly in relation to firm performance and industry conditions. Adapting the methodology of [Jenter and Kanaan \(2015\)](#), we employ a two-stage approach that decomposes firm performance into a systematic component driven by peer group performance and a firm-specific residual, and that allows the sensitivity of forced turnover to firm-specific performance to vary with local Protestant adherence and with the presence of a fixed-term agreement. This extension develops the hypothesis and the estimating framework; we leave the corresponding evidence to future drafts.

The remainder of this paper is organized as follows. Section 2 reviews the relevant literature and develops the theoretical underpinnings of how Protestant ethics can influence the choice between implicit and explicit contracting, together with our hypotheses. Section 3 describes our sample and the characteristics of the employment agreements observed in our data. Section 4 presents our main empirical evidence on the relationship between Protestant adherence and contract choice. Section 5 extends our analysis to

examine how Protestant culture relates to CEO forced turnover decisions. Finally, Section 6 concludes with a discussion of our findings and their implications for corporate governance and contracting practices.

2 Literature and hypothesis development

In the realm of corporate finance and governance, the Protestant ethic, as conceptualized by **Weber (1930)**, offers a unique lens through which to examine the influence of local culture on economic behavior and corporate practices. Central to Weber's argument is the Protestant emphasis on approaching worldly endeavors with a sense of divine purpose. By elevating work to a divinely ordained task or vocation, Protestantism fosters a disciplined and rational approach to economic activity. This perspective represents a significant departure from traditional Catholic views, which often prioritized spiritual life over secular work. The Protestant work ethic's elevation of labor to a spiritual calling has profound implications for business practices and corporate governance. It encourages individuals to view their professional activities not merely as a means to an end, but as an end in themselves - a way to fulfill their divine purpose. This mindset potentially manifests in management practices that prioritize long-term value creation, ethical behavior, and stakeholder interests, as these align with the notion of work as a calling.

In the context of contracting theory, this cultural influence becomes particularly relevant. Contract theory, as articulated by scholars such as **Klein (1996)**, examines how principals and agents construct and develop legal agreements under conditions of uncertainty and asymmetric information. The theory posits that different parties have varying incentives to perform or not perform particular actions, and contracts serve to align these incentives.

However, the Protestant ethic introduces an additional dimension to this framework. In areas with high Protestant adherence, the shared cultural emphasis on work as a calling and the importance of ethical behavior may serve as an informal governance mechanism.

This cultural backdrop could potentially reduce the need for explicit, detailed contracts between firms and their CEOs. Gillan et al. (2009) and Song and Wan (2017) have demonstrated that the choice between explicit and implicit contracting is influenced by factors such as business environment uncertainty, anticipated firm growth, and CEO characteristics. Building on their work, we propose that the level of Protestant adherence in a firm's location adds another crucial factor to this decision-making process. The disciplined and rational approach to economic activity fostered by the Protestant ethic may reduce concerns about opportunistic behavior. Both CEOs and firms, operating within this cultural context, may be more inclined to "do the right thing" without the need for extensive contractual stipulations. This aligns with Klein (1996)'s argument that reputational concerns ensure contract adherence, even when the final transaction date is known, and with Baker et al. (2002), who show that implicit agreements are sustainable precisely when reputational incentives are strong enough to make formal specification unnecessary.

Furthermore, the Protestant emphasis on long-term thinking and ethical behavior may reduce the perceived need for explicit incentives to encourage firm-specific human capital investment. CEOs in high Protestant adherence areas may be more likely to view their role as a long-term commitment and a means to fulfill their calling, rather than merely a job. This cultural influence on contracting practices is supported by the literature on social trust and its role in governance. Gulati (1995) shows that interfirm trust reduces the likelihood of equity-based alliances, suggesting that trust can substitute for formal governance mechanisms. Similarly, Guiso et al. (2004) find that trust plays a more crucial role in accessing financial contracts in areas with weaker legal enforcement.

Based on this synthesis of Protestant ethic, contracting theory, and social capital literature, we hypothesize that firms in areas with high Protestant adherence are less inclined to choose FEAs with their CEOs. We posit that these firms have fewer incentives to act opportunistically and may have a higher propensity to enhance long-term growth and care for stakeholder interests, reflecting the Protestant ethic of discipline, responsibility, and viewing work as a calling.

Our study aims to provide empirical evidence for this relationship, exploring how the Protestant work ethic, by elevating work to a divine calling and fostering a disciplined approach to economic activity, influences the choice CEO FEAs. This research contributes to a more nuanced understanding of how deeply-held cultural values shape corporate governance practices and contract design in modern financial markets.

3 Sample selection and variable construction

We select our sample based on several criteria. This section describes the various sources of data used throughout the paper, including the county-level religiosity data, employment agreements for the selected sample, firm and CEO characteristics, and demographics data by the chosen geography.

3.1 CEO employment contract data

Our study examines firms included in the Execucomp database after January 1, 2005. We utilize a comprehensive dataset of CEO employment agreements collected from the Securities and Exchange Commission (SEC) filing system, exploiting the mandatory nature of these filings for U.S.-listed firms to ensure a robust and representative sample.

Our data collection primarily spans the period from 2005 to 2022, leveraging the enhanced classification and electronic filing systems implemented by the SEC’s EDGAR system post-2005. We extract all EXHIBIT-10, EXHIBIT-99, and EXHIBIT-5 filings appended to various SEC forms, including 8-K, DEF-14A, 10-Q, 10-K, and S-4. This methodology enables us to capture a wide spectrum of documents potentially containing CEO employment agreements.

While our focus is on the post-2005 period, this choice is driven by significant data limitations prior to 2005. We observe that before 2005, many Exhibit filings were integrated with main documents, necessitating extraction from these primary filings. As illustrated in Fig 2, the number of Execucomp database firms fluctuated between 1,500 and 2,000 prior to 2005, while the number of Exhibit filings increased substantially

during this period. Consequently, we concentrate our analysis on CEOs employed after 2005 to ensure data consistency and reliability.

We retrieve 993,445 filings from the SEC website. To process them efficiently, we employ a multi-stage filtering approach. Initially, we implement a bag-of-words model to screen documents, retaining only those containing at least one keyword commonly associated with employment agreements. The initial bigrams are presented in Appendix table 2. This initial filtering step reduces our corpus to approximately 139,352 filings, a more manageable subset of potentially relevant documents. Building on recent advancements in natural language processing (Wang et al., 2023; Ashok and Lipton, 2023; Hu et al., 2022), we explore three primary methodologies for identifying employment contracts within our refined set of SEC filings:

1. In-Context Learning (ICL): Leverages the LLM’s ability to understand task-specific information provided within the input prompt.
2. Data Augmentation (DA): Involves creating synthetic examples to expand our training dataset and improve model robustness.
3. Supervised Fine-Tuning (SFT): Involves further training the LLM on a labeled dataset specific to our task.

Each method offers unique advantages: ICL provides flexibility without modifying the base model, DA addresses potential data scarcity issues, and SFT offers the possibility of achieving highly specialized performance.

In our approach to identifying employment contracts, we draw insights from Xu et al. (2023), who provide a comprehensive overview of Generative Information Extraction models. While their study indicates that In-Context Learning (ICL), Data Augmentation (DA), and Supervised Fine-Tuning (SFT) can all achieve accuracy rates of at least 85%, with SFT potentially reaching 95%, our methodology choice is primarily driven by practical considerations. A critical factor in our decision-making process is the context length limitation of language models. Given that many SEC filings exceed standard context windows, we have opted for the pre-trained models Claude Haiku 3 and Claude Sonnet

3.5. These models offer expansive context windows of 200K tokens, accommodating up to 150,000 words, which significantly mitigates the need for document segmentation. However, this choice precludes the use of SFT functionality. Consequently, we implement the ICL method, leveraging the models’ extensive context windows to maintain document coherence while benefiting from the advanced capabilities of large language models. To enhance our ICL approach, we implement two additional strategies:

1. Following [Zhang et al. \(2023\)](#), we set up the QA4RE (Question-Answering for Relation Extraction) framework, reformulating the extraction task as a series of question-answering problems.
2. We incorporate a backward verification process as described in [Weng et al. \(2022\)](#), which helps to validate and refine our initial extractions.

These enhancements aim to boost the accuracy and reliability of our contract identification and information extraction process. Utilizing our chosen LLM models, we further refined our dataset by classifying these remaining filings into 20 distinct categories,

- A) Employment Agreements
- B) Equity Incentive Plans
- C) Loan Agreements
- D) Lease Agreements
- E) Major Supplier or Customer Agreements
- F) Merger and Acquisition Related Agreements
- G) Intellectual Property Agreements
- H) Retirement and Benefit Plans
- I) Corporate Governance Documents
- J) Independence Related Agreements

- K) Guarantee and Indemnification Agreements
- L) Restructuring and Divestiture Agreements
- M) Dispute Resolution Agreements
- N) Environmental and Compliance Related Agreements
- O) Financial Advisory Agreements
- P) Change in Control Agreement
- Q) Severance Agreement
- R) Severance and Separation Agreement
- S) Separation Agreement
- T) Service Agreement

After our initial classification, we conducted a more rigorous validation process to ensure the accuracy of our employment agreement categorization. We observed that our initial setup was less likely to misclassify employment contracts into other categories. However, there was a possibility of other agreements, such as Severance Agreements, Consulting Agreements, or Change in Control Agreements, being incorrectly classified as Employment Agreements. To address this, we employed the more advanced Claude Sonnet 3.5 model to re-analyze our remaining filings. We specifically emphasized the distinctions between Employment Agreements and other types such as Retirement Agreements, Change in Control Agreements, Severance Agreements, and Separation Agreements.

Following this automated re-classification, we manually checked each filing for accuracy. Drawing on the methodology of [Gillan et al. \(2009\)](#), we established a criterion that agreements containing at least two provisions from among change in control, separation of service, or compensation plans would be retained in our final dataset. This multi-step validation process ensured a more precise and reliable classification of CEO employment

agreements for our subsequent analysis. After this process, 6,373 explicit employment contracts (including 4,660 FEAs) between January 1, 2005 and December 31, 2022, covering 3,468 CEOs and 2,068 firms remained for this study.

It is important to note that our data collection procedure differs significantly from those employed by Gillan et al. (2009) and Song and Wan (2017). While these studies primarily relied on firms' proxy statements to identify the existence of employment agreements before searching for the actual contracts, our approach is more comprehensive and direct. We observed that firms often have varying interpretations of what constitutes an employment agreement, leading to potential discrepancies in self-reporting. This inconsistency is prevalent even among S&P 500 firms. For instance, Campbell Soup Company (CIK 16732) stated in its 2019 proxy statement that it does not have employment agreements with its CEO or other executive officers. However, our search revealed a letter agreement with CEO Mark A. Clouse¹. This phenomenon is not rare, as we encountered similar instances with other firms (e.g., CIK 60667, 27419, 51252, 310158).

Conversely, some firms claim to have employment agreements, but upon inspection, these are actually letter agreements (e.g., CIK 764478, 24545, 37996, 202058, 833444; see Appendix table 3). Additionally, approximately 15% of firms do not report if having employment agreements with their Named Executive Officers (NEOs) in their proxy statements. Gillan et al. (2009) assumes these firms do not have employment agreements. Yet our search uncovered letter agreements for these companies (e.g., CIK 814453, 55067)².

These documents, often termed "letter agreements," typically cover comprehensive contractual elements such as base salary, long-term incentive plans, signing-on payments, change in control provisions, severance benefit plans, and severance policies. Despite their comprehensive nature, firms may not classify them as employment contracts due to differing interpretations. Under such circumstances, our approach which directly

¹available at: <https://www.sec.gov/Archives/edgar/data/16732/000001673218000090/ex10offerletter.htm>

²available at: <https://www.sec.gov/Archives/edgar/data/814453/000119312519206923/d767832dex101.htm> and <https://www.sec.gov/Archives/edgar/data/55067/000162828017009485/k9282017ex101.htm>

searches and classifies SEC filings allows us to capture these agreements that might be overlooked or misclassified in firms’ self-reporting, thereby providing a more accurate and comprehensive dataset of CEO employment contracts. Figure 1 summarizes the full data-processing procedure, and Appendix table 1 reproduces representative proxy-statement language while Appendix table 3 lists the specific firms whose disclosures we were able to contradict from their own filings.

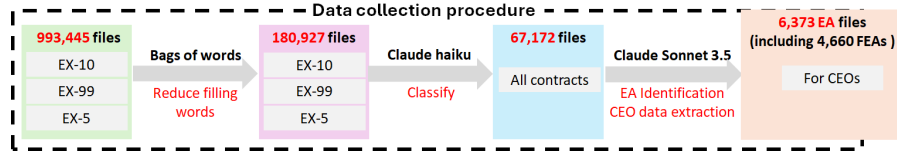


Figure 1: Contract data processing procedure

3.2 Religiosity data

Our study employs data on religiosity and religious composition from the American Religion Data Archive (ARDA), a comprehensive dataset covering 302 religious groups that participated in at least one of the 1980-2020 data collections. This dataset provides detailed information on the number of churches, members, adherents, and the religious tradition category for each group at the county level. Our primary variable of interest is the degree of Protestant adherence in the county where each sample firm is headquartered, which we use as a proxy for the firm’s exposure to Protestant social norms.

We calculate the Protestant adherence rate as the ratio of Evangelical and Mainline Protestant adherents in the county (as reported by ARDA) to the total county population (obtained from the U.S. Census Bureau). The ARDA surveys, conducted in 1990, 2000, 2010, and 2020, serve as the foundation for our religiosity measures. For non-survey years, we linearly interpolate the degree of Protestant adherence between survey years and extrapolate for the years after 2020. This approach allows us to construct a continuous measure of Protestant influence for our entire sample period.

In addition to Protestant adherence, we also examine total religiosity rates and Catholic adherence rates to provide a more comprehensive understanding of the religious

landscape's impact on corporate decisions. By utilizing this rich dataset on religious composition, we can explore how variations in local religious norms across different denominations might influence CEO employment contract design and execution.

3.3 County-level data

County-level demographic data is from the U.S. Census Bureau. We use data from the most recent four census years (1990, 2000, 2010 and 2020) and linearly interpolate and extrapolate for non-survey years. We use the standard controls from the literature, which affect the local religiosity of a county. The variables are log of total population of the county; income, measured as the per capita personal income; education, which is the fraction of the county population which is above 25 years of age and have completed at least a bachelors' degree; male-female ratio, measured as the ratio of number of males to the number of females in the county; and minority as the fraction of minority population in the county's population.

3.4 Firm-level data

Following [Gillan et al. \(2009\)](#) and [Song and Wan \(2017\)](#), we incorporate a comprehensive set of control variables to account for firm characteristics that may influence CEO employment agreements. We include return on assets (ROA), calculated as net income before extraordinary items scaled by total assets, to measure firm profitability ([Cziraki and Jenter, 2018](#)). To capture sustained poor performance, we employ a binary variable, Loss, which equals 1 if the net income before extraordinary items is negative in both the current and prior fiscal year, and 0 otherwise ([Jiang et al., 2018](#)).

To proxy for financial leverage and capital structure complexity, we use Leverage, defined as the sum of long-term and current debt divided by market value ([Song and Wan, 2017](#)). We also include a dummy variable, Havingsubdebt, which equals 1 if the firm has subordinated debt, and 0 otherwise, following.

To account for firm growth, we employ Sale_Change, the yearoveryear percentage change in sales for each company, approximated by the difference in log sales ([Jiang](#)

et al., 2018). We also include Ln_Sale, the natural logarithm of sales, to control for size effects (Gillan et al., 2009).

Investment intensity is captured by Inv, the ratio of capital expenditures to tangible long-term assets (property, plant, and equipment) (Song and Wan, 2017). To control for firm-specific risk, we include Ret_Volatility, measured as the volatility of stock returns over the fiscal year. Lastly, we incorporate market-related factors: Value_Weight_Return, the value-weighted return of the stock's industry based on the Fama-French 48 industry classification, and Company_Return, the stock return over the fiscal year (Jiang et al., 2018). These variables capture industry-wide trends and firm-specific market performance, respectively.

3.5 CEO characteristics data

To account for individual-level factors that may influence our outcomes of interest, we utilize CEO characteristics from the Execucomp database. This comprehensive set of variables allows us to control for heterogeneity in executive compensation, incentive structures, and personal attributes that could potentially impact the design and implementation of employment agreements.

We measure CEO Total Compensation as the sum of annual salary, bonus, long-term incentive plan payouts, other annual compensation, stock options granted (valued using the Black-Scholes model), restricted stock granted (at market value), and all other compensation. This comprehensive measure captures the full scope of executive remuneration. To account for the structure of CEO incentives, we include the Equity Ratio, defined as the proportion of equity-based compensation (comprising restricted stocks and option grants) to total compensation.

To distinguish between internal promotions and external hires, we employ the Outside CEO variable, a binary indicator set to 1 if the CEO joined the firm less than one year prior to their appointment. This variable allows us to control for potential differences in bargaining power and firm-specific knowledge between internally promoted and externally hired CEOs. Furthermore, we control for CEO gender in the forced-turnover analysis of

Section 5 to address potential heterogeneity in religious beliefs and preferences among executives.

3.6 CEO Forced turnover data

To analyze CEO forced turnover events, we utilize comprehensive datasets covering the period from 1993 to 2020, of which the years overlapping our contract sample (2005 to 2020) enter the analysis. Specifically, we employ two widely-recognized sources in the executive turnover literature. For the period 2001-2020, we rely on the dataset compiled by [Peters and Wagner \(2014\)](#), which provides detailed information on CEO turnover events for a broad sample of U.S. public firms. For the earlier period of 1993-2000, we incorporate data from [Jenter and Kanaan \(2015\)](#), who meticulously classified CEO turnover events for a large sample of firms. Both datasets employ similar methodologies to identify forced turnover events, primarily based on press releases and news articles surrounding CEO departures. Following the approach outlined in [Parrino \(1997\)](#), these studies classify a turnover as forced if the CEO is reported to have been fired, forced out, or left due to unspecified policy differences. Additionally, cases where CEOs below age 60 resign suddenly without another position lined up or health-related reasons are typically categorized as forced departures. We obtained these datasets from Florian Peters' website (<https://www.florianpeters.org/data/>), which provides a centralized repository for this valuable turnover information. By combining these two sources, we create a comprehensive and consistent record of forced CEO turnover events. Because our CEO contract data begin in 2005 and the turnover data end in 2020, the forced-turnover analysis in Section 5 is estimated over 2005–2020.

4 Effects of religiosity on choice of FEAs

4.1 Descriptive statistics

Table 2 and Table 3 present the evolution of CEO employment agreement adoption from 2005 to 2022 across different segments of U.S. public companies in the Execucomp database, focusing on CEOs who began their tenure after 2005. Because the sample builds up as post-2005 appointments accumulate, the firm counts for 2005 and 2006 cover only a fraction of the Execucomp universe, and the earliest years should be read with that in mind. We categorize firms into three groups: all Execucomp firms, S&P 500 companies, and non-S&P 500 companies within the Execucomp universe. Table 2 following the methodology of Gillan et al. (2009) focuses on explicit employment contracts. For each year and category, it reports the total number of firms, the number of firms with CEO contracts, and the percentage of firms with contracts. The data reveal that the overall percentage of firms with explicit CEO contracts has remained relatively stable, fluctuating around 50-53% for all Execucomp firms. Notably, S&P 500 companies consistently exhibit a lower percentage of firms with explicit contracts compared to non-S&P 500 companies.

Table 3 focuses on fixed term employment agreements and presents similar data pattern. In contrast to explicit contracts, the adoption of FEAs shows a declining trend over time, particularly in recent years. The percentage of all Execucomp firms with FEAs decreased from 35.50% in 2005 to 26.15% in 2022. This trend is observed across all firm categories, suggesting a broader shift in contracting practices. The observed patterns in contract adoption across different firm segments and over time provide valuable insights into the evolving nature of CEO employment agreements. These findings extend our understanding of the factors influencing contract choice, as discussed by Jiang et al. (2018), and highlight the need for further investigation into the determinants of these contracting trends.

Table 2 and Table 3 here

Table 4 presents comprehensive statistics on key compensation provisions stipulated

in CEO FEAs. Our sample is categorized into three groups: whole sample, S&P 500, and non-S&P 500 firms, allowing for a better understanding of FEA usage across different market segments over time. While our results are broadly consistent with previous studies by Gillan et al. (2009) and Song and Wan (2017), direct comparisons should be made cautiously due to differences in sample composition and time periods. For instance, Gillan et al. (2009) focused on explicit employment agreement for 494 S&P 500 firms in 2000, while Song and Wan (2017) examined firms in the S&P 500 index from 1993 to 2008. Our study mainly focus on FEAs and spans from 2005 to 2022, capturing more recent trends in CEO contracting practices.

Insert Table 4 here

Despite differences in sample composition, some contract provisions show similarities across studies. For example, the prevalence of base salary specification in our sample (96.37%) aligns closely with Gillan et al. (2009) (91%) and Song and Wan (2017) (96%). Similarly, the mean initial contract length in our study (37.34 months) is comparable to the 43.85 months reported by Song and Wan (2017). Notably, our data reveals differences between S&P 500 and non-S&P 500 firms. In terms of compensation (Panel A), S&P 500 firms offer higher initial salaries (mean \$1.27 million vs. \$0.78 million) and larger target bonuses (1.76 vs. 1.16 times salary) compared to non-S&P 500 firms. Regarding contract terms (Panel B), S&P 500 firms tend to have longer initial employment periods. Other provisions (Panel C) also differ: S&P 500 firms are more likely to address pension plans in their employment contracts (26.51% vs. 11.46%).

Table 5 presents a comprehensive analysis of firms with and without CEO fixed-term employment agreements (FEAs), offering insights into the determinants and implications of these contracts. The data, divided into county-level, firm-level, and CEO-level variables, reveals several significant differences between firms that use FEAs and those that do not. At the county level, firms in areas with higher Protestant rates are less likely to use FEAs (MeanDiff = 0.015***), aligning with theories on the influence of local culture on corporate governance. Interestingly, the total religious adherence rate (TotRate) shows no significant difference, while the Catholic adherence rate (CathRate)

is higher in areas with FEAs (MeanDiff = -0.018***), suggesting a potential different denominational effect on governance practices.

Firm-level variables provide further insights. Companies with FEAs tend to have lower `ln_assets` (MeanDiff = 0.325***), suggesting smaller firms are more inclined to use FEAs. These firms also exhibit higher leverage (MeanDiff = -0.250**) and lower sales (MeanDiff = 0.306***), indicating a potential relationship between financial structure, performance, and the use of FEAs. Notably, firms with FEAs have a higher incidence of subordinated debt (`Having_Subdebt`, MeanDiff = -0.032***) and higher losses (`Loss`, MeanDiff = -0.025**).

In the CEO-level variables, we particularly note the `Outside_CEO` variable. The significant negative difference (MeanDiff = -0.101***) indicates that firms with outside CEOs are more likely to have FEAs. This result strongly aligns with the findings of Gillan et al. (2009) and supports traditional contracting theory. It suggests that companies hiring external CEOs are more inclined to use FEAs, likely to mitigate information asymmetry and align incentives in situations where the CEO's background and working style may be less familiar to the firm.

These findings collectively support the hypothesis that FEAs are more common in smaller firms and those facing greater uncertainty, such as when hiring external CEOs or navigating financial complexities. The data underscores the role of FEAs as a governance mechanism to align interests and manage risk in situations where traditional informal arrangements may be insufficient.

Insert Table 5 here

4.2 Protestant adherence rates and FEAs

This study examines the relationship between local religious culture, as proxied by county-level Protestant adherence rates, and the propensity for explicit CEO contracting. We extend beyond univariate analysis to test whether this relationship persists after controlling for other potential determinants. Our empirical strategy employs probit

regression models to predict the effect of religiosity on the presence of fixed-term CEO employment agreements (FEAs). The baseline specification is:

$$\text{FEA}_{i,t} = \alpha + \beta \text{ProtestantRate}_{k,t} + \gamma \text{FirmLevelControls}_{i,(t-1)} + \delta \text{CEOLevelControls}_{i,t} + \sigma \text{CountyLevelControls}_{k,t} + \text{Year}_t + \text{Industry}_j + \varepsilon_{i,k,t} \quad (1)$$

where i , j , k , and t index firm, industry, county, and year, respectively. The dependent variable is binary, indicating the presence of an FEA. The key independent variable is *ProtestantRate*, measuring the Protestant adherence rate in the firm's headquarter county. Following Hilary and Hui (2009), we include county-level demographic controls to mitigate omitted variable bias stemming from potential correlations between religiosity and other location-specific characteristics relevant to executive contracting. These controls include log(population), education, income, male-female ratio, and minority population percentage. Consistent with the explicit-implicit contracting literature (Gillan et al., 2009; Song and Wan, 2017), we incorporate firm-level controls (lagged by one year to address potential reverse causality) and CEO characteristics. Firm controls include log(assets), leverage, book-to-market ratio, ROA, loss indicator, and presence of subordinated debt. CEO controls encompass total compensation, the equity ratio, age, share ownership, option holdings, and outsider status. All regressions employ heteroskedasticity-robust standard errors clustered at the firm and industry level. Variable definitions are provided in Table 1.

First three columns in Table 6 present our regression analyses examining the relationship between Protestant adherence rates in a firm's headquarter county and the incidence of FEAs. We employ a series of probit models with progressively more controls, ranging from a baseline model with *ProtestantRate* alone and no fixed effects to a comprehensive model including all firm, county, and CEO characteristics, as well as year and industry fixed effects. Consistent with Gillan et al. (2009), the positive and significant coefficients for outside CEO status support the notion that CEOs with higher

firm-specific human capital investment prefer explicit contracts to mitigate appropriation risk. Across all specifications, the coefficient on *ProtestantRate* is negative and statistically significant at the 1% level, supporting our hypothesis that firms in counties with higher Protestant adherence rates are less likely to use CEO FEAs. The effect is economically as well as statistically significant; estimates from our full model indicate that a one standard deviation increase in the Protestant adherence rate from its mean is associated with an economically meaningful decrease in the probability of using a CEO FEA, after controlling for all relevant characteristics. This effect remains robust, with alternative specifications showing similar magnitudes. These findings suggest that local Protestant culture significantly influences the CEO-firm contracting relationship, with firms in more Protestant areas appearing to have more governed pay practices, potentially relying more on implicit contracts upheld by goodwill market mechanisms. Our results align with and extend previous literature on determinants of CEO contracting.

Columns (4) to (6) and (7) to (9) of Table 6 present analyses using two additional measures of contract explicitness to validate our main findings. The first measure, 'Explicit_Index', quantifies contract explicitness based on key compensation provisions in CEO employment agreements. For this measure, we employ an ordered logistic regression. The second measure, 'Ln_Text_Length', is the natural logarithm of the contract's text length, based on the assumption that longer contracts tend to describe rights and duties more comprehensively, indicating higher explicitness. We use Ordinary Least Squares (OLS) regression for this analysis. The results from both measures are consistent with our primary findings on the choice between explicit and implicit contracts. The negative and statistically significant coefficients across all specifications indicate that certain factors consistently lead to less explicit contracts.

These findings further support our hypothesis that local religious culture, along with firm-specific characteristics, significantly influences the explicitness of CEO employment contracts. Specifically, we find that employment agreements tend to be less explicit when firms are characterized by higher Protestant adherence rates in their headquarter counties. This reinforces the importance of considering cultural and firm-specific factors

in corporate governance research, particularly in the context of CEO contract design.

Insert Table 6 here

4.3 FEA choice: total religiosity versus Catholic adherence

Table 7 presents the effects of total religious rate (TotRate) and Catholic rate (CathRate) on CEO Fixed Employment Agreement (FEA) choice. This examination is crucial in understanding the differential impacts of various religious adherence rates on corporate governance practices, particularly in the context of CEO contracting.

The regression results reveal a lack of statistically significant relationships between TotRate and the likelihood of CEO FEAs. Across all specifications (columns 1-3), the coefficients for TotRate are not statistically significant at conventional levels. This lack of significance persists even when controlling for firm-level and CEO-level characteristics, as well as when incorporating year and industry fixed effects. In contrast, the results for CathRate (columns 4-6) show a strong, positive, and statistically significant relationship with the likelihood of CEO FEAs. The coefficients for CathRate are consistently positive and significant at the 1% level across all specifications, ranging from 0.875 to 0.975. This relationship remains robust even after controlling for firm-level and CEO-level characteristics, as well as including year and industry fixed effects.

These findings present an interesting contrast to previous results regarding Protestant adherence rates. While Protestant rates were negatively associated with FEA use, Catholic rates show a positive relationship. This differential impact between Catholic and Protestant ethics in shaping corporate governance practices challenges the straightforward application of Weber's Protestant work ethic theory in this context. The significant positive effect of Catholic adherence rates on FEA likelihood suggests that Catholic cultural influence may encourage more formalized contractual relationships in corporate settings. This could be interpreted as a preference for explicit, codified agreements in Catholic-influenced corporate cultures, possibly reflecting a different approach to trust and contractual relationships compared to Protestant-influenced environments.

The insignificant results for TotRate suggest that general religiosity, when not differentiated by specific doctrinal emphases, may not sufficiently capture the cultural nuances that influence corporate contracting decisions. The contrasting findings between CathRate and previous results on Protestant rates highlight the importance of considering specific religious denominations, rather than overall religiosity, when examining the influence of cultural factors on corporate governance practices. These results underscore the complex relationship between religious cultural influences and corporate governance, suggesting that different religious traditions may shape business practices and contractual relationships within firms in distinct ways. Further research may be needed to fully understand the mechanisms through which Catholic and Protestant ethics differentially influence CEO contracting practices.

Insert Table 7 here

5 CEO forced turnover

In this section, we explore the potential mitigating effect of Protestant culture on CEO forced turnover probability, particularly in the context of firm performance and industry conditions.

5.1 Theoretical Background and Hypothesis Development

Traditional economic theory suggests that corporate boards primarily assess CEO performance based on firm-specific outcomes to determine retention or dismissal (Holmström (1979), Gibbons and Murphy (1992)). Firm-specific performance, typically measured as the idiosyncratic component of stock returns unexplained by market or industry factors, is crucial in evaluating a CEO's contribution to firm value.

We hypothesize, however, that shared Protestant culture and trust within a firm may significantly influence this evaluation process, particularly in high-uncertainty environments. This hypothesis builds on Weber (1930) seminal work on Protestant ethics and economic activity, as well as recent studies demonstrating the impact of religious and

cultural factors on corporate governance and decision-making (Hilary and Hui (2009), McGuire et al. (2012)). Specifically, we posit that firms with higher Protestant adherence rates may exhibit greater leniency in assessing firm-specific performance, especially when operating under conditions of heightened uncertainty.

Central to our hypothesis is the notion that boards in firms with higher Protestant adherence may interpret firm-specific performance more leniently due to shared values of perseverance and long-term orientation. This cultural alignment could foster a more nuanced understanding of the challenges CEOs face in generating idiosyncratic returns, particularly in volatile markets. We argue that these boards are more likely to attribute poor firm-specific performance to exogenous factors or temporary setbacks rather than to managerial incompetence or lack of effort. This perspective aligns with the Protestant ethic of viewing work as a vocation and emphasizing long-term commitment over short-term results.

In firms operating under high uncertainty, often characterized by the presence of explicit CEO contracts, we posit that boards may be even more inclined to exercise patience in their evaluations of firm-specific performance. Shared Protestant values could lead to a more contextual interpretation of poor idiosyncratic outcomes, recognizing that such results may not necessarily reflect managerial capabilities or dedication. This cultural lens could potentially reduce the sensitivity of forced turnover to poor firm-specific performance, as boards may be more inclined to view these challenges as opportunities for organizational learning rather than grounds for dismissal.

Conversely, for firms operating in low-uncertainty environments, where market conditions are generally favorable, we expect the impact of Protestant culture on turnover decisions based on firm-specific performance to be less pronounced. These firms typically rely on implicit rather than explicit CEO employment agreements, reflecting established norms and expectations. In such cases, boards may be less forgiving of underperformance in idiosyncratic measures, even in the presence of shared cultural values, as the expectation for success is higher given the more stable operating environment. The absence of explicit contracts in these firms underscores a presumption of consistent firm-specific

performance and may lead to stricter evaluations of CEO effectiveness, regardless of cultural affiliations.

5.2 Empirical strategy

In our empirical strategy, we employ a two-stage regression approach, adapting the methodology of [Jenter and Kanaan \(2015\)](#) to incorporate the effects of Protestant culture and explicit contracts on CEO turnover sensitivity to performance. The first stage of our analysis decomposes firm performance into a systematic component caused by peer group performance and firm-specific components:

$$r_{i,t-1} = \beta_0 + \beta_1 \cdot r_{\text{peer},t-1} + \gamma_{i,t-1} \quad (2)$$

where:

- $r_{i,t-1}$ is the firm's stock return in year $t - 1$
- $r_{\text{peer},t-1}$ is the peer group stock performance in year $t - 1$
- $\gamma_{i,t-1}$ is the residual firm-specific performance

We use the [Fama and French \(1997\)](#) classification of 48 industries to define industry returns, which serve as the peer group performance. This first stage allows us to decompose firm performance into a systematic component caused by peer group performance and firm-specific components. In the second stage, we predict the probability of forced CEO turnover using an interaction of Protestant adherence rates, contract presence, and

firm-specific performance:

$$\begin{aligned}
Probability(CEOdismissal_{i,t}) = & \alpha_0 + \alpha_1 \cdot ProtestantRate_{i,t-1} \cdot FEA_i \cdot \gamma_{i,t-1} \\
& + \alpha_2 \cdot ProtestantRate_{i,t-1} \cdot FEA_i \\
& + \alpha_3 \cdot FEA_i \cdot \gamma_{i,t-1} \\
& + \alpha_4 \cdot ProtestantRate_{i,t-1} \cdot \gamma_{i,t-1} \\
& + \beta_1 \cdot ProtestantRate_{i,t-1} \cdot FEA_i \cdot Vol_{i,t-1} \\
& + \beta_2 \cdot FEA_i \cdot Vol_{i,t-1} \\
& + \beta_3 \cdot ProtestantRate_{i,t-1} \cdot Vol_{i,t-1} \\
& + \beta_4 \cdot Vol_{i,t-1} \\
& + \alpha_5 \cdot \gamma_{i,t-1} \\
& + \alpha_6 \cdot FEA_i \\
& + \alpha_7 \cdot ProtestantRate_{i,t-1} \\
& + \alpha_8 \cdot X_{i,t} + \epsilon_{i,t}
\end{aligned} \tag{3}$$

where:

- $ProtestantRate_i$ is the Protestant adherence rate in the firm's location
- FEA_i is a binary variable indicating the presence of an FEA
- $\gamma_{i,t-1}$ is the firm-specific performance measured by stock return and industry adjusted return
- $Vol_{i,t-1}$ is the stock volatility
- $X_{i,t}$ represents a vector of control variables
- $\epsilon_{i,t}$ is the error term

This two-stage procedure effectively implements an instrumental variables estimation, with peer group performance serving as the instrument for firm performance. In this model, $\hat{\gamma}$ represents the estimated firm-specific performance component. The coefficient

α_5 captures the main effect of firm-specific performance on CEO turnover probability. Given that firm-specific performance is partially determined by CEO skill, we anticipate $\alpha_5 < 0$, indicating an inverse relationship between firm-specific performance and the likelihood of CEO dismissal, and our forbearance hypothesis predicts $\alpha_1 > 0$.

6 Conclusion

This paper asks whether local religious culture shapes the most basic formal instrument of the principal–agent relation: the written employment agreement between a firm and its chief executive. Weber (1930)’s account of the Protestant ethic carries a direct contracting implication. If Protestant norms of discipline, personal responsibility, and work as a calling make opportunistic behavior less likely on both sides of the relation, then the parties have less that they need to codify. We relate county-level Protestant adherence to whether a firm and its CEO sign a fixed-term employment agreement (FEA) and, conditional on an agreement existing, to how explicit that agreement is.

Answering the question required building the data. Rather than infer the existence of agreements from proxy-statement language — which we show to be inconsistent even among S&P 500 issuers — we go directly to the contracts. We retrieve 993,445 exhibit filings from EDGAR, screen them with a bag-of-words filter, and use large language models under an in-context-learning design to classify the surviving documents and extract their terms, followed by a second-pass reclassification with a stronger model and manual verification. The resulting sample contains 6,373 explicit CEO employment contracts, of which 4,660 are FEAs, covering 3,468 CEOs at 2,068 Execucomp firms between January 1, 2005 and December 31, 2022. The effort is warranted: firms that deny having CEO employment agreements in their proxy statements not infrequently have letter agreements, complete with salary, severance, and change-in-control terms, sitting in their own exhibit filings. The descriptive picture that emerges is that the share of Execucomp firms with an explicit CEO contract has been stable at roughly 50 to 53 percent, while fixed-term agreements have receded steadily, from 35.50 percent of

firms in 2005 to 26.15 percent in 2022. The decline is concentrated among large firms: S&P 500 FEA use falls from 36.89 percent in 2006 to 18.50 percent in 2022, whereas non-S&P 500 firms decline only from 37.40 percent to 29.16 percent and remain the heavier users of both fixed-term and explicit contracting throughout.

Local Protestantism matters for that choice. Firms without FEAs are headquartered in counties whose mean Protestant adherence rate is 0.211, against 0.196 for firms with FEAs, and the relation survives county, firm, and CEO controls together with year and industry fixed effects. The estimated effect of Protestant adherence on the probability of an FEA ranges from -0.701 to -1.165 and is significant at the one percent level in every specification. The same pattern holds on the intensive margin: Protestant adherence loads negatively on the Explicit Index built from ten key compensation provisions (-1.619 to -1.596) and on the log length of the contract text (-4.673 to -4.225), both at the one percent level. Where local Protestant adherence is higher, CEO agreements are not merely less often fixed-term; they are also less detailed and shorter.

The effect is denominational rather than a proxy for religiosity in general. Total adherence rates are statistically indistinguishable from zero in every specification, whereas Catholic adherence enters positively and significantly at the one percent level, with coefficients between 0.875 and 0.975. We do not present this as a clean placebo. Local Catholicism appears to push firms toward codified, fixed-term arrangements just as local Protestantism pushes them away, which is difficult to reconcile with a story based on generic religiosity or on risk aversion, and which qualifies any simple application of Weber's thesis: the operative distinction in our data is between Protestant and Catholic norms, not between the religious and the secular.

Section 5 sets out a second question that our data make it possible to ask: whether these cultural forces reach the ex post governance decision. Following [Jenter and Kanaan \(2015\)](#), we decompose firm returns into a component explained by [Fama and French \(1997\)](#) 48-industry peer performance and a firm-specific residual, and specify a model in which the sensitivity of forced CEO turnover to that residual varies with local Protestant adherence and with the presence of a fixed-term agreement. If shared norms of persever-

ance and long-term orientation lead boards to attribute poor idiosyncratic performance to exogenous factors rather than to managerial failure, the triple interaction of Protestant adherence, FEA status, and firm-specific performance should be positive. We develop that hypothesis and the estimating framework here and leave the corresponding empirical evidence to future work.

Taken together, the results support a substitution view of informal and formal governance. Contracting theory explains explicit agreements as protection for firm-specific human capital under uncertainty, and our controls behave accordingly: outside CEOs hold more, and more detailed, written contracts. Our contribution is to show that the demand for that protection is conditioned by the social environment surrounding the firm's headquarters, and that the conditioning is denomination-specific. The findings also caution against reading contractual formality as a measure of governance quality: fewer and shorter written terms in high-Protestant counties may reflect a substitution of informal for formal enforcement rather than weaker oversight.

Three limitations bound the interpretation. First, we proxy for the culture in which a manager operates using the religious composition of the headquarters county, which captures the firm's social environment rather than the beliefs of any individual CEO or director, and firms choose where to locate. Second, we exploit cross-sectional and slowly moving variation in adherence without an exogenous shock, so our estimates are conditional associations rather than causal effects; we cannot rule out an omitted local factor that drives both adherence and contracting practice. Third, our contract data are machine-extracted, so residual classification and extraction error remains despite second-pass validation and manual review. Headquarters relocations and CEO moves across counties offer within-firm and within-manager variation that would sharpen identification, executive-level measures of religiosity would sharpen the mechanism, and the natural next question — whether the implicit contracting we document is efficient *ex post* — is one that our data now make it possible to ask.

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Figure 2: Exhibit filings and Execucomp Firms

The blue line illustrates the number of Exhibit filings from the SEC. Prior to 2005, some of these Exhibit filings were typically integrated with main documents, but for this analysis, they were extracted from primary corporate filings when necessary. The trend shows a significant increase in Exhibit filings beginning in the early 2000s, reaching its peak around 2005-2006, followed by a gradual decline. In contrast, the orange line represents the number of firms included in the Execucomp database. This metric demonstrates relative stability, fluctuating between 1,500 and 2,000 companies in the period leading up to 2005.

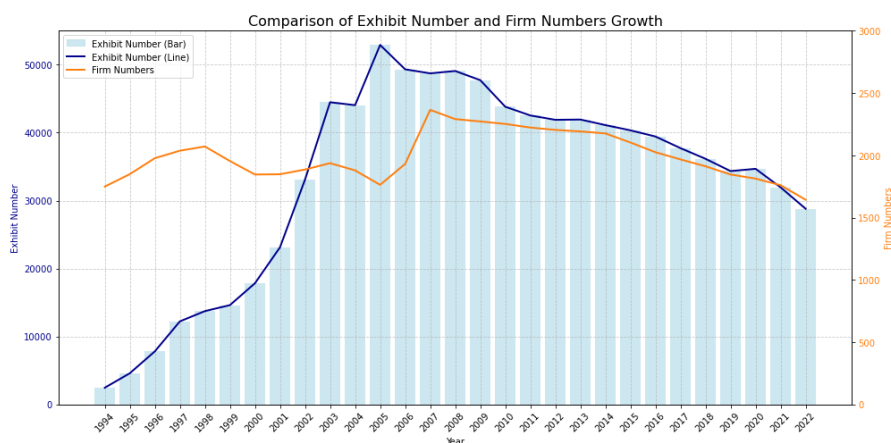
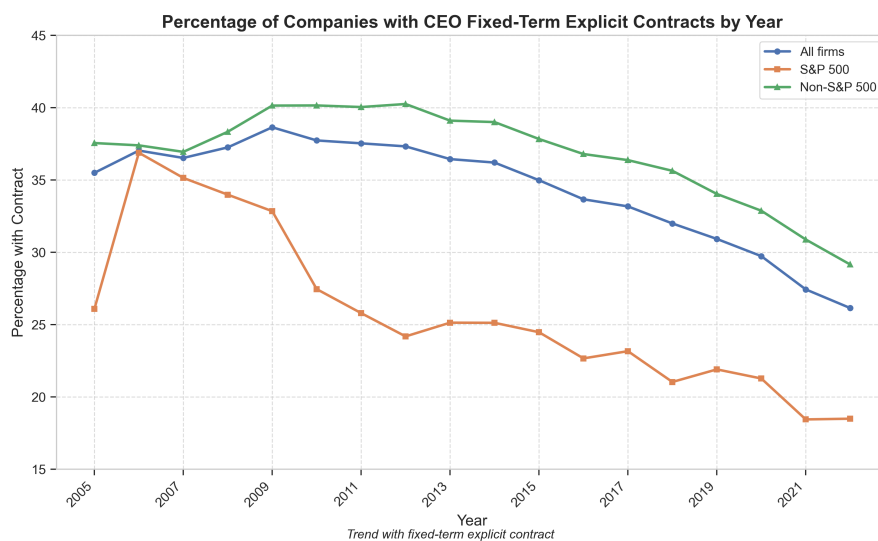


Figure 3: Dynamics in CEO Fixed-Term Employment Agreements

Figure 3 illustrates the percentage of companies with CEO fixed-term employment agreements from 2005 to 2022, categorized into three groups: All firms, S&P 500 firms, and Non-S&P 500 firms.



Trend with FEAs

Table 1: Variable definitions

Variable Name	Variable definition
<i>Contracting Variable</i>	
FEA Contract	Fixed term employment agreement Binary variable equal to one if the CEO has an explicit employment agreement in the compensation year, and zero otherwise
Ln_Text_Length	Natural logarithm of the length of each CEO's contract
Explicit_Index	Measure of contract explicitness based on ten key compensation provisions in the CEO employment agreement. The explicit index sums the presence of the following provisions: Base salary specification; Target bonus clause; Maximum bonus stipulation; Restricted stock grant provision; Contract duration terms; Change in control clause; Confidentiality agreement; Non-compete clause; Termination for good cause provision; Resignation for good reason clause.
<i>County-level Variables</i>	
ProtestantRate	Ratio of the number of Evangelical and Mainline Protestant adherents in the county (as reported by ARDA) to the total population of the county (as reported by the Census Bureau)
CathRate	Ratio of the number of Catholic adherents in the county (as reported by ARDA) to the total population of the county (as reported by the Census Bureau)
TotRate	Ratio of the number of religious adherents (as reported by ARDA) to the total population of the county (as reported by the Census Bureau)
LnPop	Natural logarithm of a county's population
LnIncome	Natural logarithm of a county's per capita personal income
Edu (%)	The percentage of 25 years or older residents in a county who have completed at least a bachelor's degree
Male_Female	The ratio of male population to female population in a county
Minor	The percentage of Hispanic, African American or other non-white residents in the total population
<i>CEO Characteristic Variables</i>	
Outside_CEO	Indicator variable that equals 1 if the CEO is appointed in that position within 1 years of joining the firm, and zero otherwise
Age	Age of CEO in the compensation year
Equity_ratio	The ratio of equity-based compensation (restricted stock and stock option grants) to total compensation for the year prior to the compensation year.

Continued on next page

Table 1 – continued from previous page

Variable Name	Variable definition
Total_Compensation	Sum of base salary, bonus, long-term incentive payouts, other annual compensation, stock options granted (Black-Scholes value), restricted stocks granted (market value), and all other compensation earned in the compensation year
Own	Fraction of the company's shares owned by the CEO
Ophold	Dollar value of options held by the CEO
Tenure	The duration of the executive's CEO tenure
<i>Firm Variables</i>	
Ln_assets	Natural logarithm of total assets
Book_to_market	The ratio of book value of equity to market value for the year
Roa	Net income before extraordinary items/total assets.
Loss	Dummy variable that equals 1 if the net income before extraordinary items is negative in the current and prior fiscal year, and 0 otherwise.
Havingsubdebt	Dummy variable that equals 1 if the firm has subordinated debt, and 0 otherwise
Leverage	The sum of longterm and current debt divided by market value
Ln_Sale	Natural logarithm of sale
Sale_Change	Year-over-year percentage change in sales for each company, approximated by the difference in log sales
Sale_Change_Volatility	The standard deviation of Sale_Change over the preceding fiscal years
Inv	The ratio of capital expenditures to tangible long-term assets (property, plant, and equipment)
Ret_Volatility	The volatility of stock returns over the fiscal year
Value_Weight_Return	The value-weighted return of the stock's industry, based on the Fama-French 48 industry classification
Company_Return	The stock return over the fiscal year
<i>Forced Turnover Variables</i>	
Forced Turnover	Indicator variable that equals 1 if the CEO departure in the year is classified as forced, and zero otherwise, following Parrino (1997)
Gamma (γ)	Firm-specific performance: the residual from a first-stage regression of the firm's stock return on the return of its Fama and French (1997) 48-industry peer group in year $t - 1$
Vol	The volatility of the firm's daily stock returns over year $t - 1$

Table 2: Distribution of Explicit Employment Agreements

Table 2 presents the evolution of explicit employment contract adoption from 2005 to 2022 across different segments of U.S. public companies included in the Execucomp database. The data is categorized into three groups: all Execucomp firms, S&P 500 companies, and non-S&P 500 companies within the Execucomp universe. For each category, the table shows the total number of firms, the number of firms with CEO contracts, and the percentage of firms with contracts. The S&P 500 and non-S&P 500 columns do not sum to the whole-sample column; the residual consists of firm-years belonging to neither subgroup.

YEAR	Total firms	Total contract	Total % with contract	S&P 500 total	S&P 500 contract	S&P 500 % with contract	Non-S&P 500	Non-S&P 500 contract	Non-S&P 500 % with contract
2005	307	151	49.19	69	26	37.68	221	117	52.94
2006	648	332	51.23	122	59	48.36	500	263	52.60
2007	1325	663	50.04	246	93	46.04	1008	558	53.29
2008	1460	765	52.40	253	120	47.43	1166	630	54.03
2009	1550	830	53.55	277	127	45.85	1238	692	55.90
2010	1611	858	53.26	295	129	43.73	1285	716	55.72
2011	1670	887	53.11	344	138	41.40	1321	739	55.94
2012	1736	927	53.40	339	141	41.59	1366	767	56.15
2013	1797	949	52.81	362	151	41.71	1409	784	55.64
2014	1831	962	52.54	374	153	40.91	1428	796	55.74
2015	1815	943	51.96	384	154	40.10	1406	780	55.48
2016	1779	919	51.66	397	153	38.54	1367	760	55.60
2017	1769	930	52.57	410	168	40.98	1344	759	56.47
2018	1747	910	52.09	423	177	41.84	1313	731	55.67
2019	1707	876	51.32	429	182	42.42	1269	692	54.53
2020	1688	874	51.78	437	193	44.16	1241	679	54.71
2021	1662	849	51.08	439	189	43.05	1214	660	54.37
2022	1560	776	49.74	427	182	42.62	1125	593	52.71

Table 3: Distribution of FEAs

Table 3 presents the evolution of FEA adoption from 2005 to 2022 across different segments of U.S. public companies included in the Execucomp database. The data is categorized into three groups: all Execucomp firms, S&P 500 companies, and non-S&P 500 companies within the Execucomp universe. For each category, the table shows the total number of firms, the number of firms whose CEO has a fixed-term employment agreement, and the corresponding percentage. The S&P 500 and non-S&P 500 columns do not sum to the whole-sample column; the residual consists of firm-years belonging to neither subgroup.

YEAR	Whole sample	Total Fixed EA	Total % with Fixed EA	S&P 500	S&P 500 Fixed EA	S&P 500 % with Fixed EA	Non-S&P 500	Non-S&P 500 Fixed EA	Non-S&P 500 % with Fixed EA
2005	307	109	35.50	69	18	26.09	221	83	37.56
2006	648	240	37.04	122	45	36.89	500	187	37.40
2007	1325	484	36.53	262	71	35.15	1008	402	36.95
2008	1460	544	37.26	253	86	33.99	1166	447	38.34
2009	1550	599	38.65	277	91	32.85	1238	497	40.15
2010	1611	608	37.74	295	81	27.46	1285	516	40.16
2011	1670	627	37.54	314	81	25.80	1321	529	40.05
2012	1736	648	37.33	339	82	24.19	1366	550	40.26
2013	1797	655	36.45	362	91	25.14	1409	551	39.11
2014	1831	663	36.21	374	94	25.13	1428	557	39.01
2015	1815	635	34.99	384	94	24.48	1406	532	37.84
2016	1779	599	33.67	397	90	22.67	1367	503	36.80
2017	1769	587	33.18	410	95	23.17	1344	489	36.38
2018	1747	559	32.00	423	89	21.04	1313	468	35.64
2019	1707	528	30.93	429	94	21.91	1269	432	34.04
2020	1688	502	29.74	437	93	21.28	1241	408	32.88
2021	1662	456	27.44	439	81	18.45	1214	375	30.89
2022	1560	408	26.15	427	79	18.50	1125	328	29.16

Table 4: Major provisions of CEO employment agreements

Table 4 presents comprehensive data on CEO FEA terms across different segments of the U.S. stock market. The data is categorized into three groups: the whole sample, S&P 500 companies, and non-S&P 500 companies. For each category, the table shows various contract elements divided into three panels: Panel A (Salary and Bonus), Panel B (Contract Terms), and Panel C (Other Provisions). Each panel provides information on specific aspects of FEA, including percentages of FEA containing certain provisions, as well as mean values where applicable. The S&P 500 and non-S&P 500 columns do not partition the whole sample, because index membership is unavailable for a residual group of contracts.

Index	All		S&P500		Non S&P500	
	%	Mean	%	Mean	%	Mean
Panel A: Salary and Bonus						
Initial salary(\$ mil)	96.37%	0.85	93.49%	1.27	96.73%	0.78
Target bonus (multiple of salary)	66.69%	1.25	91.07%	1.76	64.43%	1.16
Cash signing bonus (multiple of salary)	17.36%	1.06	15.98%	1.78	17.37%	0.96
Initial restricted stock grant (\$ mil)	25.02%	3.53	33.73%	6.35	24.00%	2.92
Panel B: Contract Terms						
Change control protection (months)	51.14%	21.06	46.15%	22.88	52.47%	20.00
Change control payment(multiple of salary)	53.67%	2.53	50.89%	2.59	54.01%	2.52
Non compete period (months)	87.69%	19.12	87.57%	18.85	87.77%	19.05
Contract length at initial terms (months)	100%	37.34	100%	40.74	100%	36.64
Termination no cause payment (multiple of salary)	75.61%	1.93	72.19%	2.05	76.10%	1.91
Panel C: Other Provisions						
Insider trading	6.60%	6.60%	10.24%	10.24%	6.20%	6.20%
Change control	86.41%	86.41%	89.16%	89.16%	86.10%	86.10%
Confidential	96.82%	96.82%	96.99%	96.99%	96.71%	96.71%
Non compete	93.16%	93.16%	92.77%	92.77%	93.15%	93.15%
Term good cause	99.05%	99.05%	99.40%	99.40%	98.97%	98.97%
Resign good reason	95.95%	95.95%	98.19%	98.19%	95.59%	95.59%
Clawback	31.16%	31.16%	36.75%	36.75%	30.70%	30.70%
Due to MA	5.88%	5.88%	8.43%	8.43%	4.98%	4.98%
Pension plan	13.67%	13.67%	26.51%	26.51%	11.46%	11.46%

Table 5: Descriptive Statistics for Firms With and Without CEO Fixed-Term Employment Agreements

Table 5 presents descriptive statistics for variables related to CEO fixed-term employment agreements. The data is divided into three main panels: Panel A (County-level Variables), Panel B (Firm-level Variables), and Panel C (CEO-level Variables). For each variable, the table shows the number of observations (N) and the mean for two subsamples: firm-years without an FEA (FEA==0) and firm-years with an FEA (FEA==1). The Diff column reports the difference in means between the two subsamples (No FEA minus FEA) and the final column the p-value of the associated test, with asterisks indicating statistical significance levels.

	No FEA		FEA		Diff (No FEA - FEA)	p-Value
Variables	N	Mean	N	Mean		
Panel A: County-level Variables						
ProtestantRate	2905	0.211	1266	0.196	0.015	0.000***
TotRate	2905	0.509	1266	0.509	-0.000	0.966
CathRate	2905	0.222	1266	0.240	-0.018	0.000***
Male_Femal	2907	0.961	1267	0.959	0.002	0.035**
LnPop	2907	13.811	1267	13.907	-0.096	0.000***
LnIncome	2907	10.465	1267	10.437	0.027	0.003***
Edu	2907	0.273	1267	0.234	0.039	0.000***
Minor	2907	0.556	1267	0.521	0.035	0.006***
Panel B: Firm-level Variables						
Book_to_Market	2418	1.325	1123	1.204	0.121	0.321
Having_Subdebt	2778	0.067	1222	0.099	-0.032	0.000***
Loss	2778	0.156	1222	0.181	-0.025	0.049**
Roa	2735	-0.002	1211	0.002	-0.004	0.653
Ln_assets	2738	7.590	1213	7.265	0.325	0.000***
Leverage	2418	0.810	1123	1.060	-0.250	0.011**
Sale	2719	7.121	1204	6.815	0.306	0.000***
Sale_Change	2553	0.065	1164	0.064	0.001	0.913
Sale_Change_Vol	2455	0.183	1150	0.193	-0.01	0.273
Inv	2606	0.254	1128	0.145	0.009	0.378
Ret_Volatility	2471	0.115	1105	0.118	-0.003	0.380
Value_Weight_Return	2472	0.124	1107	0.108	0.016	0.093*
Company_Return	2472	0.182	1107	0.149	0.033	0.571
Panel C: CEO-level Variables						
Age	2894	53.208	1262	52.966	0.242	0.301
Total_Compensation	2877	5.614	1254	5.675	-0.061	0.858
Equity_Ratio	2808	0.461	1227	0.438	0.023	0.024**
Outside_CEO	2907	0.556	1267	0.657	-0.101	0.000***
Own	2638	0.802	1186	1.030	-0.228	0.384
Ophold	2891	2.782	1263	2.414	0.368	0.440
Tenure	2179	4.221	1071	4.937	-0.716	0.0000***

Table 6: Results of Predicting Use of FEA

Table 6 presents regression results examining the determinants of CEO FEA use and of contract explicitness. The table has nine columns arranged in three blocks of three. Columns (1) to (3) are probit models in which the dependent variable is a binary indicator for the presence of an FEA; marginal effects are reported. Columns (4) to (6) are ordered logit models using *Explicit_Index*, a measure of contract explicitness based on ten key compensation provisions in the CEO employment agreement. The *Explicit index* sums the presence of the following provisions: Base salary specification, Target bonus clause, Maximum bonus stipulation, Restricted stock grant provision, Contract duration terms, Change in control clause, Confidentiality agreement, Non-compete clause, Termination for good cause provision, and Resignation for good reason clause. Columns (7) to (9) are OLS regressions using the natural logarithm of the length of each CEO's contract text as the dependent variable. Columns (4) to (9) are estimated on the same sample as columns (1) to (3); *Explicit_Index* and *Ln_Text_Length* are set to zero for firm-years in which the CEO has no employment agreement.

	FEA			Explicit_Index			Ln_Text_Length		
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)
ProtestantRate	-0.701*** (-3.884)	-0.983*** (-4.027)	-1.165*** (-3.611)	-1.619*** (-6.354)	-1.604*** (-4.637)	-1.596*** (-3.497)	-4.673*** (-6.853)	-4.505*** (-5.296)	-4.225*** (-4.074)
Male_Femal			-2.170** (-2.233)			-2.850** (-2.083)			-3.956 (-1.197)
LnPop			0.058* (1.777)			0.058 (1.279)			0.127 (1.145)
LnIncome			-0.235 (-1.592)			-0.066 (-0.343)			-0.136 (-0.276)
Edu			-0.126 (-0.310)			-0.273 (-0.509)			-0.448 (-0.319)
Minor			-0.306*** (-3.362)			-0.305*** (-2.662)			-0.689*** (-2.608)
Book_to_Market			0.021 (0.753)			0.006 (0.184)			0.068 (0.724)

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Table 6 continued

	FEA			Explicit_Index			Ln_Text_Length		
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)
Having_Subdebt			0.201*			0.210			0.304
			(1.902)			(1.306)			(0.789)
Loss			-0.020			-0.041			-0.036
			(-0.215)			(-0.321)			(-0.111)
Roa			0.124			0.046			-0.241
			(0.739)			(0.172)			(-0.383)
Ln_Assets			-0.120**			-0.164***			-0.468***
			(-2.502)			(-2.644)			(-2.817)
Leverage			0.029*			0.020			0.067
			(1.847)			(0.849)			(1.070)
Sale			0.027			0.058			0.123
			(0.564)			(0.920)			(0.734)
Sale_Change_Volatility			0.057			-0.027			0.162
			(0.417)			(-0.147)			(0.312)
Inv			-0.352**			-0.353**			-0.498
			(-2.191)			(-1.961)			(-0.954)
Age			-0.003			-0.020***			-0.033**
			(-0.771)			(-3.492)			(-2.292)
Total_Compensation			0.004			0.012*			0.040***
			(1.395)			(1.768)			(3.314)
Equity_Ratio			-0.071			0.501***			1.074***
			(-0.690)			(3.479)			(2.951)
Outside_CEO			0.274***			1.264***			2.532***
			(4.797)			(14.831)			(12.418)

Continued on next page

Table 6 continued

	FEA			Explicit_Index			Ln_Text_Length		
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)
Own			-0.003 (-0.542)			-0.031 (-0.840)			-0.038 (-1.186)
Opthold			0.001 (0.411)			-0.003 (-0.904)			-0.005 (-0.712)
Ret_Volatility			-0.048 (-0.121)			-0.064 (-0.093)			-1.158 (-0.743)
Value_Weighted_Return			-0.034 (-0.304)			-0.185 (-1.161)			-0.378 (-1.049)
Company_Return			0.014 (0.462)			0.029 (0.640)			0.017 (0.153)
Constant	-0.371*** (-8.852)	0.281 (0.599)	4.791** (2.242)				6.268*** (38.745)	3.298*** (2.782)	8.348 (1.237)
Year FE	No	Yes	Yes	No	Yes	Yes	No	Yes	Yes
Industry FE	No	Yes	Yes	No	Yes	Yes	No	Yes	Yes
Observations	4171	4047	2970	4171	4059	2980	4171	4059	2980
Pseudo- R^2 / R^2	0.003	0.046	0.073	0.003	0.020	0.063	0.011	0.057	0.156

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Table 7: Effects of Total Religious Rate and Catholic Rate on CEO FEA Choice

Table 7 presents regression results examining the effects of the total religious adherence rate (TotRate) and the Catholic adherence rate (CathRate) on CEO FEA choice. The dependent variable in all columns is FEA, a binary variable equal to one if the CEO has a fixed-term employment agreement in the compensation year. Columns (1) to (3) show the results for TotRate, while columns (4) to (6) show the results for CathRate. County-, firm- and CEO-level controls are included only in columns (3) and (6), as indicated at the foot of the table. T-statistics of the coefficient estimates are reported in parentheses below the estimates, with standard errors clustered by firm and industry. All variables are defined in Table 1. Statistical significance levels are indicated as follows: * 10% level, ** 5% level, *** 1% level.

	FEA (1)	FEA (2)	FEA (3)	FEA (4)	FEA (5)	FEA (6)
TotRate	0.009 (0.043)	-0.084 (-0.308)	-0.309 (-0.951)			
CathRate				0.875*** (4.751)	0.975*** (3.996)	0.960*** (3.047)
Firm-level controls	NO	No	Yes	No	No	Yes
CEO-level controls	NO	No	Yes	No	No	Yes
County-level controls	NO	No	Yes	No	No	Yes
Observations	4171	4047	2970	4171	4047	2970
Year Fixed Effects	No	Yes	Yes	No	Yes	Yes
Industry Fixed Effects	No	Yes	Yes	No	Yes	Yes
R-sq	0.000	0.041	0.068	0.004	0.046	0.071

Appendix table 1: Excerpts from Proxy Statements

This table reports excerpts from proxy statements, together with firm name, report date, and text expressing the kind of agreement with executives.

Firm Name	Date of report	Text from DEF14A
WELLTOWER INC.	2022-04-08	Welltower does not have employment agreements with Mr. McHugh, Mr. Burkart, Mr. McQueen or Ms. Menon. Welltower's policy is not to enter into any employment agreements in the future with any employee other than the CEO.
HUNTINGTON BANCSHARES	2021-03-12	Only one executive officer, our CEO, has an employment agreement with us, which is described under Mr. Steinour's Employment Agreement below.
LENNAR CORP	2023-03-01	We do not have employment agreements or change in control agreements with any of our NEOs or other executive officers. This gives the Compensation Committee flexibility to change the components of our executive compensation program in order to remain competitive in the market and address economic conditions.
JACOBS SOLUTIONS INC	2022-03-12	There are no employment agreements in effect with any of our NEOs except for Mr. Hill. Mr. Hill has an employment agreement with the Company because of our standard practice to enter into employment agreements with Australian employees.
VIATRIS INC.	2023-03-01	No New fixed-term NEO employment agreements. The Viatrix Compensation Committee and Board determined that, other than Mr. Coury's previously disclosed 2020 employment agreement, the Board will not approve any new fixed-term NEO employment agreements going forward.
SEALED AIR CORP	2022-04-14	We sometimes enter into offer letter agreements with our executives in connection with their recruitment. We entered into an offer letter agreement, dated September 5, 2017, with Mr. Doheny in connection with his recruitment, which was amended on December 10, 2020 and August 30, 2022.
UNITEDHEALTH GROUP INC	2022-04-22	We have entered into an employment agreement with each of the named executive officers.

Appendix table 2: Initial Bigrams for Searching Employment Agreement Terms

This table presents a representative sample of bigrams used in our document screening process to identify potential employment agreements. It's important to note that these terms are not an exhaustive list, but rather examples of the regular expressions we implemented in our bag-of-words model. We used regular expressions based on these and many more related terms to conduct a comprehensive screening of documents. This approach allowed for flexible matching, capturing variations and related phrases beyond those explicitly listed in the table. For instance, a regular expression might capture plurals, different word orders, or additional words between the key terms. The table is organized into three columns, each containing phrase patterns commonly associated with employment contracts and related documents. By using regular expressions based on these patterns, we were able to rapidly identify filings that potentially contained employment contracts, significantly streamlining the process of locating relevant agreements within a large corpus of corporate filings. This method provided a robust foundation for our initial document screening, allowing for efficient narrowing of the document pool for subsequent, more detailed analysis. The flexibility of regular expressions enabled us to cast a wide net while maintaining precision in our search for relevant employment agreement terms.

employment agreement	employee agreement	employer agreement
employment contract	employee contract	employer contract
employment terms	employee terms	employer terms
letter of agreement	letter agreement	employment offer
employee offer	employer offer	employment termination
employee termination	employer termination	offer letter
letter of appointment	offer of employment	employment letter agreement
employee letter agreement	employer letter agreement	new position
new assignment	base salary	

Appendix table 3: Examples of Firms with Discrepancies in CEO Employment Contract Disclosures

This table presents examples of firms where there are discrepancies between the CEO employment contract information disclosed in the proxy statement and the actual contract terms found in offer letters or other SEC filings. The table includes the following information for each firm: index number, Execucomp identifier (EXECID), company name (CONAME), Central Index Key (CIK), URL for the proxy statement, and URL for the offer letter or other document containing the actual contract terms.

Index	EXECID	CONAME	CIK	Proxy Statement	Offer Letter Address
1	46069	CAMPBELL SOUP CO	16732	https://www.sec.gov/Archives/edgar/data/16732/000120677419003416/cpb3618521-def14a.htm	https://www.sec.gov/Archives/edgar/data/16732/000001673218000090/ex10offerletter.htm
2	28093	TARGET CORP	27419	https://www.sec.gov/Archives/edgar/data/27419/000130817915000165/lgt2015_def14a.htm	https://www.sec.gov/Archives/edgar/data/27419/000002741914000028/tgt-20140802xexhibit10cc.htm
3	35380	LOWE'S COS INC	60667	https://www.sec.gov/Archives/edgar/data/60667/000119312519110021/d687530ddef14a.htm	https://www.sec.gov/Archives/edgar/data/60667/000119312518170222/d577212dex101.htm
4	51370	BEST BUY CO INC	764478	https://www.sec.gov/Archives/edgar/data/764478/000114036120009075/nc10007505x1_pre14a.htm	https://www.sec.gov/Archives/edgar/data/764478/000076447819000016/bby-20190413xex10_2.htm
5	35321	FORD MOTOR CO	37996	https://www.sec.gov/Archives/edgar/data/37996/000003799620000071/f09302020exhibit101.htm	https://www.sec.gov/Archives/edgar/data/37996/000003799620000071/f09302020exhibit101.htm
6	34834	JOHNSON CONTROLS INTL PLC	833444	https://www.sec.gov/Archives/edgar/data/833444/000104746913000246/a2212357zdef14a.htm	https://www.sec.gov/Archives/edgar/data/833444/000104746912010665/a2211375zex-10_5.htm
7	35072	KELLOGG CO	55067	https://www.sec.gov/Archives/edgar/data/55067/000162828019002643/k2019307def14a.htm	https://www.sec.gov/Archives/edgar/data/55067/000162828017009485/k9282017ex101.htm